

# Ordinal compacta and the ball fixed point property for $C(K, p)$

arXiv automatic math run `fa_banach_001`

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## Source Question

The source paper is Aviles–Japon–Lennard–Martinez–Cervantes–Stawski, *The ball fixed point property in spaces of continuous functions*, arXiv:2506.17995 [1]. A Banach space has the ball fixed point property, abbreviated BFPP, if every nonexpansive self-map of its closed unit ball has a fixed point.

The final list of questions asks in item (i) whether there is an infinite compact  $K$  and a non-isolated  $p \in K$  such that  $C(K, p)$  has the BFPP, where  $C(K, p) = \{f \in C(K) : f(p) = 0\}$ .

REMARK 1.4. This is a much more delicate set-theoretic question. One can construct such  $P$ -points under the Continuum Hypothesis [29], but there are other models of set theory in which  $\mathbb{N}^*$  contains no  $P$ -points [35].

We finally conclude the paper by highlighting several questions that may arise after reading the article, which we believe could be of significant interest for future research in the area:

- i) Is it possible to find an infinite compact set  $K$  and a non-isolated point  $p \in K$  such that  $C(K, p) = \{f \in C(K) : f(p) = 0\}$  has the BFPP?
- ii) Is it possible to construct an explicit example of a fixed point free nonexpansive mapping from the unit ball of  $\ell_\infty/c_0$  into itself (without turning to CH)?
- iii) Is it possible to exhibit an example of a fixed point free nonexpansive mapping from the unit ball of  $C(K)$  into itself, whenever  $C(K)$  fails to be order-complete? In this case, we would have a topological characterization of the BFPP for Banach spaces of continuous functions:  $C(K)$  satisfies the BFPP if and only if  $K$  is extremally disconnected.

REFERENCES

## Idea of the Proof

For  $K = [0, \kappa]$  and  $p = \kappa$ , the space  $C(K, p)$  consists of continuous scalar functions on the compact ordinal interval that vanish at the endpoint. A one-step ordinal shift places the value 1 at 0, shifts successor values forward, leaves limit-ordinal values unchanged, and keeps the endpoint value 0. This map is nonexpansive. Any fixed point would have to be 1 at 0, hence 1 at every successor ordinal, and then by continuity 1 at every limit ordinal below  $\kappa$ . That contradicts continuity at the endpoint  $\kappa$ , where functions in  $C(K, p)$  vanish.

## Partial Result

**Theorem 1.** *Let  $\kappa$  be a nonzero limit ordinal, let  $K = [0, \kappa]$  with the order topology, and let  $p = \kappa$ . Then  $C(K, p)$  fails the ball fixed point property.*

*Proof.* Let

$$X = C(K, p) = \{f \in C([0, \kappa]) : f(\kappa) = 0\}$$

with the supremum norm. Define  $T : B_X \rightarrow B_X$  by

$$(Tf)(\alpha) = \begin{cases} 1, & \alpha = 0, \\ f(\beta), & \alpha = \beta + 1 < \kappa, \\ f(\alpha), & 0 < \alpha < \kappa \text{ is a limit ordinal}, \\ 0, & \alpha = \kappa. \end{cases}$$

First we check that  $Tf \in X$ . Successor ordinals are isolated, and 0 is isolated, so continuity is automatic there. Let  $0 < \lambda < \kappa$  be a limit ordinal. Since  $f$  is continuous at  $\lambda$ , for every  $\varepsilon > 0$  there is  $\gamma < \lambda$  such that  $|f(\alpha) - f(\lambda)| < \varepsilon$  whenever  $\gamma < \alpha \leq \lambda$ . If  $\alpha$  is a limit ordinal in  $(\gamma, \lambda]$ , then  $(Tf)(\alpha) = f(\alpha)$ ; if  $\alpha = \beta + 1 \in (\gamma + 1, \lambda)$ , then  $(Tf)(\alpha) = f(\beta)$ , and  $\gamma < \beta < \lambda$ . Thus  $(Tf)(\alpha) \rightarrow f(\lambda) = (Tf)(\lambda)$  as  $\alpha \rightarrow \lambda$ .

At the endpoint  $\kappa$ , use  $f(\kappa) = 0$ . Given  $\varepsilon > 0$ , choose  $\gamma < \kappa$  such that  $|f(\alpha)| < \varepsilon$  for  $\gamma < \alpha \leq \kappa$ . Then  $|(Tf)(\alpha)| < \varepsilon$  for all sufficiently large  $\alpha < \kappa$ , whether  $\alpha$  is a limit ordinal or a successor ordinal. Hence  $Tf$  is continuous at  $\kappa$  and  $(Tf)(\kappa) = 0$ . Also  $T(B_X) \subset B_X$ , because  $Tf$  takes only the value 1, the value 0, and values already taken by  $f$ .

The map  $T$  is nonexpansive. For  $f, g \in B_X$ , at 0 and  $\kappa$  the difference  $(Tf) - (Tg)$  is zero. At a successor  $\beta + 1 < \kappa$ , it equals  $f(\beta) - g(\beta)$ . At a nonzero limit ordinal  $\lambda < \kappa$ , it equals  $f(\lambda) - g(\lambda)$ . Therefore

$$\|Tf - Tg\|_\infty \leq \|f - g\|_\infty.$$

Finally suppose that  $f \in B_X$  is a fixed point of  $T$ . Then  $f(0) = 1$ . By transfinite induction,  $f(\alpha) = 1$  for every  $\alpha < \kappa$ : the successor step follows from

$$f(\beta + 1) = (Tf)(\beta + 1) = f(\beta),$$

and the limit step follows from continuity. Since  $\kappa$  is a limit ordinal, this forces  $\lim_{\alpha \rightarrow \kappa} f(\alpha) = 1$ , contradicting  $f(\kappa) = 0$ . Hence  $T$  is fixed-point-free, and  $X$  fails the BFPP.  $\square$

**Corollary 1.** *For every uncountable regular cardinal  $\kappa$ , the compact ordinal  $[0, \kappa]$  has a non-isolated  $P$ -point  $p = \kappa$ , yet*

$$C([0, \kappa], p)$$

*fails the BFPP.*

*Proof.* The point  $\kappa$  is non-isolated because  $\kappa$  is a limit ordinal. It is a  $P$ -point because a countable intersection of neighborhoods of  $\kappa$  contains an interval  $(\gamma, \kappa]$ , using  $\text{cf}(\kappa) > \omega$ . The failure of BFPP is Theorem 1.  $\square$

## Scope

This does not answer the source question positively or negatively in full. It rules out a large and classical family of candidates, including the standard compact ordinal with a non-isolated  $P$ -point,  $[0, \omega_1]$ . It also gives an elementary fixed-point-free map independent of the source paper's cardinal family  $K^{\kappa, \Gamma}$ .

## Novelty and verification notes

The local run indexes were searched for arXiv 2506.17995, BFPP, “ball fixed point property”,  $C(K, p)$ ,  $P$ -points, and spaces of continuous functions. No exact previous packet or attempt for this source paper was found. A bounded web search on 2026-06-14 for the same phrases found the source arXiv page and no later explicit answer to the final questions.

The only proof point needing care is continuity of  $Tf$  at limit ordinals and at  $\kappa$ . The definition of  $T$  was chosen exactly so that the successor-shift values converge to the already-prescribed limit values.

## References

- [1] A. Aviles, M. Japon, C. Lennard, G. Martinez-Cervantes, A. Stawski, *The ball fixed point property in spaces of continuous functions*, arXiv:2506.17995, 2025.